UNIVERSITY OF TECHNOLOGY FACULTY OF CSE	FINAL EXAM	Semester / Academic year	2	2024-2025
		Date		17/05/2025
	Course title	ML AND APPLICATIONS		
	Course ID	CO5241		
	Duration	90 mins	Question sheet code	4218

Notes: - Students do not use course materials except at most two A4 hand-written cheetsheets.
- Submit the question sheet together with the answer sheet.
- Choose the best answer (only 1) for each question.

The Questions 12-25 pertain to a customer churn prediction problem. The goal is to predict 'Churn' ('Yes' or 'No'). Please use the 'CustomerChurnTrain' and 'CustomerChurnTest' datasets provided below, ***where applicable***. For calculations involving logarithms, assume $\log_2$.

Training Dataset: 'CustomerChurnTrain' (10 instances)

CustomerID	SupportTickets (X1)	ServiceUsage (X2)	DiscountApplied (X3)	Churn
TR01	3	10	0	Yes
TR02	0	50	1	No
TR03	5	5	0	Yes
TR04	1	40	0	No
TR05	2	60	1	No
TR06	4	8	1	Yes
TR07	0	30	0	No
TR08	6	3	0	Yes
TR09	1	25	1	No
TR10	3	15	1	Yes

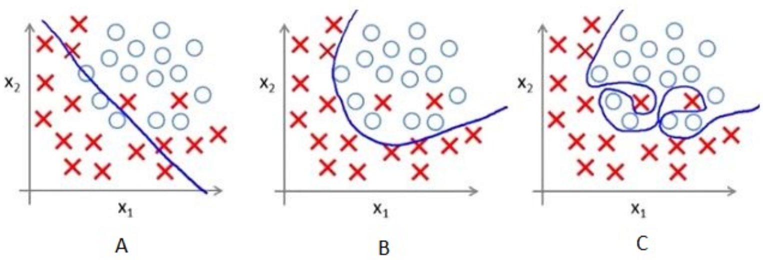
Training Summary: 5 'Yes' (Churn), 5 'No' (No Churn).

Test Dataset: 'CustomerChurnTest' (4 instances)

CustomerID	SupportTickets (X1)	ServiceUsage (X2)	DiscountApplied (X3)	Churn
TS01	4	12	0	Yes
TS02	0	55	0	No
TS03	2	20	1	No
TS04	5	6	1	Yes

Answer the Questions 12-25 pertaining to a customer churn prediction problem based on the datasets and general machine learning principles. For each question, select the best possible answer.

Questions 1–3 consider 3 figures (A, B, C) illustrating the classification boundaries of logistic regression models on a dataset.



1. **[L.O.1].** Which one presents overfitting?

A. Fig. C. B. Fig. A. C. Fig. B. D. None of them.

2. **[L.O.1].** Which of the statements below is true?

   (1) The training error of the model presented by Fig. A is maximum.
   (2) The model presented by Fig. C is the best, as the training error is minimum.
   (3) The model presented by Fig. B is the best, as it will work well with unseen data.
   (4) The performance of all models is similar.

A. (2) and (3). B. (2) and (4). C. (1) and (3). D. (1) and (2).

3. **[L.O.1].** Assume that these models were trained with different regularization terms. Which one has the maximum regularization value?

A. Fig. B. B. Fig. A. C. Fig. C. D. The same.

Questions 4 and 5 consider the model M classifying data from three classes $X, Y,$ and Z. The classification result is given by the following confusion matrix.

	Actual X	Actual Y	Actual Z
Predicted X	116	13	10
Predicted Y	14	11	20
Predicted Z	11	10	122

4. **[L.O.1].** The classification precision of model M for class X is

A. 0.832. B. 0.823. C. 0.825. D. 0.852.

5. **[L.O.1].** The classification recall of model M for class X is

A. 0.892. B. 0.289. C. 0.829. D. 0.298.

Questions 6–8 considers the following table for an exam result prediction model.

Learning hours	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

We use the logistic regression with Sigmoid function $f(z) = \frac{1}{1+e^{-z}}$. Let $z = -64 + 2*(\textit{learning hours})$, this value of z can be considered to represent the ratio of pass to fail numbers.

6. **[L.O.1, L.O.3].** For a student with 33 learning hours, the probability of passing is

A. 8%. B. 88%. C. 58%. D. 68%.

7. **[L.O.1, L.O.3].** Given a passing probability threshold of 0.95, the minimum learning hours a student needs to pass is

A. 33. B. 33.5. C. 39. D. 33.47.

8. **[L.O.1, L.O.3].** Maintaining the passing probability threshold at 0.95, the statement "if a student's learning hours are less than the minimum required learning hours, they fail the exam" is

A. True. B. False.

Questions 9–11 consider the following problem. Suppose that we have a dataset and design a linear regression model of degree-3 polynomial. We find that the training and testing error is 0 or in other words, it perfectly fits the data.

9. **[L.O.1].** What will happen when we use the degree 4 polynomial in linear regression?

A. The model should be overfit. B. The model should be underfit.
C. Cannot say. D. None of these.

10. **[L.O.1].** What will happen when we use the degree-2 polynomial in linear regression

A. The model should be overfit.
B. Cannot say.
C. The model should be underfit.
D. None of these.

11. **[L.O.1].** In terms of bias and variance, which of the following is true when we use the degree 2 polynomial?

A. Bias will be high, variance will be high.
B. Bias will be high, variance will be low.
C. Bias will be low, variance will be high.
D. Bias will be low, variance will be low.

12. **[L.O.1, L.O.3].** Using the 'CustomerChurnTest' dataset, if a model predicts 'Churn' as ['Yes', 'No', 'Yes', 'Yes'] for customers TS01, TS02, TS03, and TS04, respectively, what is the accuracy of this model on the test set?

A. 0.75
B. 0.50
C. 1.00
D. 0.25

13. **[L.O.1, L.O.3].** Based on the 'CustomerChurnTrain' dataset, what is the prior probability $P(\text{Churn} = \text{'Yes'})$?

A. 0.4
B. 0.6
C. 1.0
D. 0.5

14. **[L.O.1, L.O.3].** In the 'CustomerChurnTrain' dataset, if we split the data based on the feature 'DiscountApplied (X3)'', what is the entropy of the subset where 'DiscountApplied = 1' with respect to the 'Churn' variable? (Recall $H(S) = \sum -p_i \log_2(p_i)$).

A. 0.45
B. 0.08
C. 0.97
D. 0.52

15. **[L.O.1, L.O.3].** A Random Forest classifier, trained on a dataset similar to 'CustomerChurnTrain'', consists of 5 decision trees. For a test instance TS01, the individual trees predict 'Churn' as: [Yes, No, Yes, Yes, No]. What is the final prediction of the Random Forest for TS01?

A. No
B. Yes
C. Uncertain, needs tie-breaking
D. Yes, with probability 0.6

16. **[L.O.1, L.O.3].** A linear SVM has the decision function $f(x) = w_1x_1 + w_2x_2 + b$. Given $w_1 = 1, w_2 = -2, b = 3$. How would this SVM classify a new data point $x = (x_1 = 2, x_2 = 3)$ from a dataset similar to 'CustomerChurnTest' (e.g., x_1=SupportTickets, x_2=ServiceUsage)? (Predict class +1 if $f(x) \geq 0$, class -1 if $f(x) < 0$).

A. Class -1 (e.g., 'No Churn')
B. Class +1 (e.g., 'Churn')
C. On the decision boundary
D. Cannot be determined

17. **[L.O.1, L.O.3].** Consider two customers from the 'CustomerChurnTrain' dataset: TR01 with features (X1=3, X2=10) and TR04 with features (X1=1, X2=40). Ignoring X3, calculate the value of the linear kernel $K(TR01_{1,2}, TR04_{1,2}) = x_i^T x_j$ using only 'SupportTickets' and 'ServiceUsage''.

A. 30
B. 400
C. 43
D. 403

18. **[L.O.1, L.O.3].** An HMM has two states (S1: Active, S2: Idle) and emits observations based on website activity ('View', 'NoView'). Initial: $P(S1) = 0.7, P(S2) = 0.3$. Transitions: $P(S1|S1) = 0.8, P(S2|S1) = 0.2, P(S1|S2) = 0.4, P(S2|S2) = 0.6$. Emissions: $P(\text{View}|S1) = 0.9, P(\text{NoView}|S1) = 0.1, P(\text{View}|S2) = 0.2, P(\text{NoView}|S2) = 0.8$. What is the probability of observing the sequence $O = (\text{View})$ at the first time step $(t = 1)$?

A. 0.63
B. 0.06
C. 0.69
D. 0.9

19. **[L.O.1, L.O.3].** Suppose in an HMM training phase, after aligning observed sequences with state sequences (e.g., using Viterbi path from a previous iteration of Baum-Welch or labeled data), it's found that State S1 ('Engaged') occurred 100 times. Out of these 100 times, the observation 'Purchase' occurred 30 times. What is the re-estimated emission probability $P(\text{Purchase}|S1)$ using Maximum Likelihood Estimation?

A. 0.53
B. 0.33
C. 0.71
D. 0.45.

20. **[L.O.1, L.O.3].** A neuron in a fully connected layer has two inputs $x_1 = 2, x_2 = 3$. The weights are $w_1 = 0.5, w_2 = -1.0$, and the bias $b = 0.1$. If the activation function is ReLU ($f(z) = \max(0, z)$), what is the neuron's output?

A. 0.0
B. -1.9
C. 0.1
D. 1.0

21. **[L.O.1, L.O.3].** A simple neural network for regression is trained to predict 'ServiceUsage' (X2). For CustomerID TR01 (Actual ServiceUsage=10), the model predicts 12. For CustomerID TR03 (Actual ServiceUsage=5), the model predicts 4. What is the Mean Squared Error (MSE) for these two predictions?

A. 4.0 B. 1.0 C. 5.0 D. 2.5

22. **[L.O.1].** Why is a highly complex model (e.g., a very deep decision tree or a neural network with many parameters) more prone to overfitting the 'CustomerChurnTrain' data compared to a simpler model? What role does the size of the training dataset play in this context?

A. A complex model has high capacity (many degrees of freedom) and can memorize the training data, including its noise and specific idiosyncrasies, rather than learning the underlying generalizable patterns. A small training dataset provides insufficient information to constrain these many parameters effectively, making it easier for the model to find a complex function that fits the few examples perfectly but fails on unseen data.

B. Complex models are inherently flawed because their mathematical formulations are less stable, leading to overfitting regardless of data size. Simpler models use more robust mathematics.

C. Overfitting is only a concern for models that use iterative training methods like SGD; models with closed-form solutions (like OLS regression) do not overfit, irrespective of complexity or data size.

D. A complex model overfits because it explores too few hypotheses about the data, settling on a sub-optimal one too quickly. A larger dataset would force it to explore more hypotheses.

23. **[L.O.1].** Compare and contrast pre-pruning (early stopping) and post-pruning (e.g., cost-complexity pruning) in decision tree construction. What is a potential drawback of relying solely on pre-pruning, especially when dealing with features that only become informative in combination?

A. Pre-pruning involves removing nodes after the tree is fully grown, while post-pruning stops growth if a node has too few samples. Pre-pruning is generally more effective at preventing overfitting.

B. Post-pruning is computationally cheaper as it doesn't require growing the full tree, whereas pre-pruning evaluates many potential pruning points on the full tree.

C. Both pre-pruning and post-pruning aim to increase training set accuracy by allowing the tree to become more complex and specialized.

D. Pre-pruning stops tree growth early if splits don't meet a certain threshold (e.g., min information gain). Post-pruning grows a full tree then removes branches that don't improve generalization on a validation set. A drawback of pre-pruning is "horizon effect": a currently unpromising split might lead to very good splits further down, which pre-pruning would miss, potentially prematurely terminating growth before revealing informative feature interactions.

24. **[L.O.1].** Random Forests can provide estimates of feature importance. Describe one common method by which feature importance is calculated in a Random Forest and discuss why this metric might sometimes be misleading or biased, for instance, in the presence of highly correlated predictor variables in the 'CustomerChurnTrain' dataset.

A. Permutation Importance: Feature importance is determined by the increase in training error when that feature's values are randomly permuted. This method is generally unbiased even with correlated features.
B. Coefficient Magnitude: Similar to linear models, features with larger coefficients (weights) assigned by the trees are considered more important. This is not directly applicable to standard decision trees.
C. Mean Decrease in Impurity (MDI, e.g., Gini importance): Feature importance is calculated as the total reduction in the criterion (e.g., Gini impurity or variance) brought by that feature, averaged over all trees. It can be biased towards high-cardinality numerical features or inflate the importance of one variable in a group of correlated variables while diminishing others, as the predictive information is shared or split among them.
D. Frequency of Splits: The feature that is used for splitting most frequently across all trees is deemed the most important. This method is robust to feature correlation and cardinality.

25. **[L.O.1].** The "kernel trick" is a central concept in SVMs. Explain what the kernel trick is, and why it is computationally advantageous compared to explicitly mapping data to a high-dimensional feature space before applying a linear SVM. Illustrate with an example like trying to separate churned vs. non-churned customers if their 'ServiceUsage' and 'SupportTickets' formed a circular pattern.

A. The kernel trick involves selecting a subset of features (the kernel) that are most relevant for classification, effectively performing feature selection before training the SVM, thus reducing dimensionality and computational cost.
B. The kernel trick allows SVMs to operate in a high-dimensional feature space without explicitly computing the coordinates of the data in that space. It does this by defining a kernel function $K(x_i, x_j)$ that computes the dot product of the data points $\phi(x_i) \cdot \phi(x_j)$ in the high-dimensional space directly from the original low-dimensional inputs. This avoids the often prohibitive cost of computing and storing $\phi(x)$ if the high-dimensional space is very large or infinite.
C. The kernel trick is a method for transforming non-linear decision boundaries learned by an SVM in a high-dimensional space back into a non-linear boundary in the original feature space for easier visualization.
D. The kernel trick refers to the iterative process of finding the optimal kernel parameters (like γ for RBF) that maximize the margin, which is computationally intensive but ensures the best possible separation.

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1. A.
2. C.
3. B.
4. B.
5. C.
6. B.
7. D.
8. B.
9. A.
10. C.
11. B.
12. A.
13. D.
14. C.
15. B.
16. A.
17. D.
18. C.
19. B.
20. A.
21. D.
22. A.
23. D.
24. C.
25. B.